



Decentralized Identity on Zoo Network: Chain-Native DIDs and Verifiable Credentials via I-Chain

~~IdentityVM: Replacing External~~
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Abstract

We present I-CHAIN (IDENTITYVM), a chain-native decentralized identity layer for Zoo Network that replaces external Identity and Access Management (IAM) systems with validator-backed W3C Decentralized Identifiers (DIDs) and verifiable credentials. I-CHAIN implements the `did:lux:zoo:*` method, where validators collectively serve as identity providers through threshold credential issuance. Credentials carry zero-knowledge proofs enabling selective disclosure—proving properties (age ≥ 18 , membership in a set, credential validity) without revealing underlying data. The system supports trusted issuer registries, self-issued credentials, and on-chain revocation via accumulator-based revocation lists. Integration with T-CHAIN enables threshold signing of credentials using t -of- n validator committees, ensuring no single validator can forge identities. Formal verification of trust semantics is established through Lean 4 proofs covering authority delegation, vouching, and revocation (`Trust/Authority.lean`, `Trust/Vouch.lean`, `Trust/Revocation.lean`). We demonstrate that chain-native identity achieves equivalent security guarantees to centralized IAM while eliminating single points of failure, supporting privacy-preserving credential presentation, and enabling cross-chain identity portability across the Lux ecosystem.

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1 Introduction

1.1 The Problem with External IAM

Current blockchain ecosystems rely on external identity infrastructure. Users authenticate through centralized providers (OAuth, SAML, OIDC), creating dependencies on systems whose trust assumptions conflict with decentralization. When a centralized identity provider is compromised, all dependent services are exposed. When a provider disappears, identities become unresolvable.

Three specific failures motivate I-CHAIN:

1. **Trust Mismatch:** Centralized IAM assumes a trusted authority. Blockchain assumes trustlessness. These assumptions are incompatible.
2. **Privacy Violation:** Traditional identity systems require revealing full credentials. Proving age requires exposing a birthdate. Proving membership requires exposing an identity.
3. **Vendor Lock-in:** Identities bound to a provider cannot migrate. Users who invest in reputation on one platform lose it when switching.

1.2 Chain-Native Identity

I-CHAIN resolves these failures by making identity a first-class blockchain primitive. Every Zoo Network participant receives a `did:lux:zoo:*` identifier resolved entirely on-chain. Credentials are issued by validator committees through threshold signing, ensuring that identity provision inherits the same Byzantine fault tolerance as consensus.

1.3 Contributions

This paper makes the following contributions:

- A W3C DID method (`did:lux:zoo`) with on-chain resolution and validator-backed lifecycle management.
- A verifiable credential system with zero-knowledge selective disclosure based on Groth16 proofs.
- An accumulator-based revocation mechanism with $O(1)$ revocation checks and $O(\log n)$ witness updates.
- Formal verification of trust authority, vouch chains, and revocation correctness in Lean 4.
- Integration architecture with T-CHAIN (threshold signatures) and K-CHAIN (key management) for credential signing and key rotation.

1.4 Paper Organization

Section 2 defines the DID method specification. Section 3 covers verifiable credentials and zero-knowledge proofs. Section 4 describes the validator identity provider model. Section 5 presents credential revocation. Section 6 details T-CHAIN integration. Section 7 covers self-issued credentials. Section 8 presents formal verification. Section 9 analyzes security properties. Section 10 discusses cross-chain identity. Section 11 concludes.

2 The did:lux:zoo Method

2.1 DID Syntax

Zoo DIDs follow W3C DID Core v1.0 [1] with the method-specific identifier:

`did:lux:zoo:<network>:<identifier>`

where $\text{<network>} \in \{\text{mainnet}, \text{testnet}, \text{devnet}\}$ and <identifier> is the Base58-encoded Blake2b-256 hash of the initial public key.

Definition 1 (Zoo DID Document). A Zoo DID Document D is a tuple (id, V, S, A, M) where:

- id is the DID string,
- $V = \{vk_1, \dots, vk_m\}$ is the set of verification methods (public keys),
- $S = \{s_1, \dots, s_k\}$ is the set of service endpoints,
- $A \subseteq V$ is the authentication key set,
- M is metadata including creation time, update history, and deactivation status.

2.2 CRUD Operations

All DID operations are IDENTITYVM transactions validated by consensus:

- **Create:** Initial document signed by genesis key. Valid if DID is unused, signature verifies, and document conforms to schema.
- **Read:** Resolution queries the state trie with Merkle inclusion proofs for trustless light-client verification.
- **Update:** Signed by a key in the document's `capabilityInvocation` set. Key rotation is atomic—old key authorizes new key in a single transaction.
- **Deactivate:** Sets `deactivated` flag (irreversible). Document remains resolvable for credential verification but cannot authorize new operations.

3 Verifiable Credentials with Zero-Knowledge Proofs

3.1 Credential Structure

A verifiable credential (VC) on I-CHAIN follows the W3C Verifiable Credentials Data Model [2] with an additional ZK commitment:

Definition 2 (Zoo Verifiable Credential). A credential $C = (\text{issuer}, \text{subject}, \text{claims}, \text{proof}, \text{commitment})$ where:

- $\text{issuer} \in \mathcal{D}$ is the issuer's DID,
- $\text{subject} \in \mathcal{D}$ is the subject's DID,
- $\text{claims} = \{(k_i, v_i)\}_{i=1}^n$ are attribute key-value pairs,
- proof is the issuer's signature (threshold or individual),
- $\text{commitment} = \text{Pedersen}(v_1 \parallel \dots \parallel v_n; r)$ is a hiding commitment to claim values.

3.2 Selective Disclosure via ZK Proofs

Credential holders generate zero-knowledge proofs to disclose specific properties without revealing underlying values. We use Groth16 [3] over BN254 for proof generation.

Proposition 1 (Selective Disclosure Soundness). *Given a credential C with commitment com and a predicate ϕ over claim values, the ZK proof system Π satisfies:*

1. **Completeness:** *If $\phi(v_1, \dots, v_n) = \text{true}$, an honest prover can produce a valid proof.*
2. **Soundness:** *No PPT adversary can produce a valid proof for $\phi(v_1, \dots, v_n) = \text{false}$ except with negligible probability.*
3. **Zero-Knowledge:** *The proof reveals nothing beyond the truth of ϕ .*

Supported predicates include:

- **Range proofs:** $v_i \in [a, b]$ (e.g., $\text{age} \geq 18$ without revealing exact age).
- **Set membership:** $v_i \in S$ (e.g., $\text{country} \in \{\text{allowed jurisdictions}\}$).
- **Equality:** $v_i = c$ for a public constant c .
- **Compound:** Boolean combinations of the above.

3.3 Credential Presentation Protocol

Presentation follows a four-step flow: (1) holder computes witness $w = (v_1, \dots, v_n, r)$ and generates $\pi \leftarrow \text{Groth16.Prove}(\text{crs}, \phi, w, \chi)$; (2) holder sends $(\text{com}, \pi, \text{issuer}, \chi)$ to verifier; (3) verifier checks $\text{Groth16.Verify}(\text{vk}, \phi, \text{com}, \pi, \chi)$ and resolves the issuer DID; (4) verifier checks revocation status via the accumulator.

4 Validators as Identity Providers

4.1 Validator Identity Committee

In I-CHAIN, validators collectively serve as the identity provider. No single validator can issue credentials unilaterally. Instead, a committee of n validators is selected per epoch, and credential issuance requires threshold agreement from t of n members.

Definition 3 (Identity Committee). An identity committee $\mathcal{C}_e$ for epoch e is a subset of validators $\{v_1, \dots, v_n\} \subset \mathcal{V}$ selected by stake-weighted sortition. The committee threshold is $t = \lfloor 2n/3 \rfloor + 1$.

This ensures that credential issuance inherits the same Byzantine fault tolerance as consensus: an adversary must control $> 1/3$ of staked weight to forge a credential.

4.2 Credential Issuance Flow

1. **Request:** Subject submits a credential request $\text{req} = (\text{subject_did}, \text{claim_type}, \text{evidence})$ to the IDENTITYVM.
2. **Verification:** Each committee member v_i independently verifies the evidence against the claim type's verification policy.

3. **Partial Signature:** If verified, v_i produces a partial threshold signature σ_i over the credential body.
4. **Aggregation:** Once t partial signatures are collected, the IDENTITYVM aggregates them into a full credential signature $\sigma = \text{Aggregate}(\sigma_1, \dots, \sigma_t)$.
5. **Issuance:** The credential is stored on-chain (commitment only; claim values remain with the subject) and an issuance event is emitted.

4.3 Trusted Issuer Registry

Not all credentials require validator committee issuance. I-CHAIN maintains a *Trusted Issuer Registry* (TIR) mapping credential types to authorized issuers (DID list, minimum trust level, verification policy). Adding or removing trusted issuers requires DAO governance approval with 66% supermajority, consistent with Zoo's governance framework [7].

5 Credential Revocation

5.1 Accumulator-Based Revocation

I-CHAIN uses cryptographic accumulators [4] for efficient revocation. An accumulator acc is a short commitment to a set S of non-revoked credential identifiers.

Definition 4 (Revocation Accumulator). Let $\mathbb{G}$ be a bilinear group of prime order q . The accumulator for non-revoked set $S = \{c_1, \dots, c_m\}$ is:

$$\text{acc}_S = g^{\prod_{i=1}^m (s+c_i)}$$

where s is the trapdoor known to the committee and g is a generator.

Theorem 2 (Revocation Check Complexity). *Given a non-membership witness w_c for credential c , verifying that c has been revoked (i.e., $c \notin S$) requires $O(1)$ bilinear pairing operations.*

Proof. The non-membership witness is (d, w) where $d \cdot (s + c) + r = \prod_{c_i \in S} (s + c_i)$ for some r . Verification checks $e(\text{acc}, g) = e(w, g^{s+c}) \cdot e(g^r, g)$ using two pairings and one multiplication. $\square$

5.2 Revocation by Credential Type

Revocation authority depends on credential type:

Credential Type	Revocation Authority	Mechanism
Validator-issued	Validator committee (t -of- n)	Accumulator update
Trusted issuer	Issuing entity	Signed revocation tx
Self-issued	Subject (self)	Document deactivation
DAO-governed	DAO vote (66% supermajority)	Governance proposal

Table 1: Revocation authorities by credential type

5.3 Revocation Propagation

When a credential is revoked, the accumulator is updated atomically in the next block. Light clients that cache accumulator values receive update proofs via the `IDENTITYVM` event log. Witness updates for remaining non-revoked credentials require $O(\log n)$ computation per holder, amortized across epochs.

6 Integration with T-Chain

6.1 Threshold Credential Signing

I-CHAIN delegates cryptographic signing to T-CHAIN (Zoo’s threshold signature chain) [5]. This separation ensures that `IDENTITYVM` handles identity semantics while T-CHAIN handles key material.

The integration follows a request-response pattern:

1. `IDENTITYVM` constructs the credential body and submits a signing request to T-CHAIN.
2. T-CHAIN coordinates t -of- n partial signature generation across the validator committee.
3. T-CHAIN returns the aggregated signature to `IDENTITYVM`.
4. `IDENTITYVM` attaches the signature and finalizes the credential.

6.2 Key Rotation via K-Chain

When validator keys are rotated (managed by K-CHAIN [6]), I-CHAIN must update its committee configuration. This is handled through cross-chain events:

1. K-CHAIN emits a `KeyRotated` event with the new public key share.
2. `IDENTITYVM` updates the committee member’s verification key.
3. Outstanding signing requests using the old key are voided and resubmitted.

The formal relationship between chains is:

$$\text{I-Chain} \xrightarrow{\text{signing requests}} \text{T-Chain} \xrightarrow{\text{key material}} \text{K-Chain} \quad (1)$$

7 Self-Issued Credentials

7.1 Motivation

Not all identity claims require external verification. A user may self-assert preferences, pseudonymous handles, or application-specific attributes. I-CHAIN supports self-issued credentials that are signed by the subject’s own DID keys.

7.2 Trust Levels

I-CHAIN defines a trust hierarchy for credentials:

Verifiers specify minimum trust levels in their verification policies. A DeFi protocol might require Level 3 for KYC credentials but accept Level 0 for display names.

Level	Type	Description
0	Self-issued	Signed by subject only. No external verification.
1	Peer-vouched	Endorsed by k peers with Level ≥ 1 credentials.
2	Trusted issuer	Issued by a TIR-registered entity.
3	Validator-issued	Threshold-signed by validator committee.

Table 2: Credential trust levels

7.3 Vouch Chains

Level 1 (peer-vouched) credentials use a vouching mechanism formalized in `Trust/Vouch.lean`:

Definition 5 (Vouch Chain). A vouch chain for credential c is a directed acyclic graph $G = (V, E)$ where each node is a DID and each edge (d_i, d_j) represents d_i vouching for d_j 's claim. The credential reaches Level 1 when $|\{d_i : (d_i, \text{subject}(c)) \in E\}| \geq k$.

8 Formal Verification

8.1 Lean 4 Proof Architecture

The trust semantics of I-CHAIN are formally verified in Lean 4 across three modules:

Trust/Authority.lean — Proves that authority delegation is transitive but bounded. A DID with authority level ℓ can delegate to another DID at level $\leq \ell - 1$. This prevents authority inflation.

Theorem 3 (Authority Non-Inflation). *For any chain of authority delegations $d_0 \rightarrow d_1 \rightarrow \dots \rightarrow d_k$, the authority level is strictly decreasing:*

$$\text{auth}(d_0) > \text{auth}(d_1) > \dots > \text{auth}(d_k)$$

Trust/Vouch.lean — Proves that the vouching system is Sybil-resistant under the assumption that at most $f < k/2$ of the vouching DIDs are controlled by a single entity. The proof establishes that forging a Level 1 credential requires controlling $\geq k/2 + 1$ independent DIDs.

Theorem 4 (Vouch Sybil Resistance). *If an adversary controls at most f DIDs where $f < \lfloor k/2 \rfloor + 1$, then the adversary cannot produce a valid vouch chain for a Level 1 credential.*

Trust/Revocation.lean — Proves that revocation is:

1. **Immediate:** A revoked credential fails verification in the next block.
2. **Irreversible:** Once revoked, a credential cannot be un-revoked (the accumulator update is one-way).
3. **Non-interfering:** Revoking one credential does not affect the validity of others.

Module	Property	Status
Trust/Authority.lean	Non-inflation	Verified
Trust/Authority.lean	Bounded delegation depth	Verified
Trust/Vouch.lean	Sybil resistance	Verified
Trust/Vouch.lean	Convergence (finite vouching)	Verified
Trust/Revocation.lean	Immediate revocation	Verified
Trust/Revocation.lean	Irreversibility	Verified
Trust/Revocation.lean	Non-interference	Verified

Table 3: Formal verification status

8.2 Verification Coverage

9 Security Analysis

9.1 Threat Model

We consider an adversary $\mathcal{A}$ who controls up to $f < n/3$ validators (consistent with BFT consensus) and an arbitrary number of non-validator DIDs.

Theorem 5 (Credential Forgery Resistance). *Under the threshold signature security assumption (Definition in [5]) and the BFT quorum requirement $t = \lfloor 2n/3 \rfloor + 1$, the adversary $\mathcal{A}$ cannot forge a validator-issued credential.*

Proof. Forging requires t partial signatures. The adversary controls at most $f < n/3 < t$ validators, which is insufficient to reach the threshold $t = \lfloor 2n/3 \rfloor + 1$. $\square$

9.2 Privacy Properties

Proposition 6 (Credential Unlinkability). *Two ZK presentations of the same credential to different verifiers are computationally unlinkable, assuming the zero-knowledge property of Groth16 and fresh randomness per presentation.*

The accumulator-based revocation system reveals only whether a credential is revoked or not—it does not reveal which other credentials exist, the total number of non-revoked credentials, or the identity of the revoker.

10 Cross-Chain Identity

10.1 Identity Portability

Zoo DIDs are resolvable across the Lux ecosystem via the `did:lux:*` method family, as specified in the Lux DID specification [8]. A credential issued on Zoo can be presented and verified on any Lux-compatible chain by:

1. Resolving the issuer DID via cross-chain state proofs.
2. Verifying the credential signature against the issuer’s public keys.
3. Checking revocation status via the Zoo accumulator (relayed as a cross-chain header).

10.2 IAM Compatibility

For systems that still use traditional IAM (as in Hanzo’s OIDC-based authentication [9]), I-CHAIN provides a bridge: DID-authenticated sessions can mint short-lived OIDC tokens via a gateway, enabling backward compatibility without compromising decentralization.

11 Conclusion

I-CHAIN demonstrates that decentralized identity can replace centralized IAM without sacrificing usability or security. By making identity a chain-native primitive backed by validator threshold signing, Zoo Network achieves:

- **Decentralized trust:** No single authority controls identity issuance.
- **Privacy by default:** ZK proofs enable selective disclosure as the standard presentation mode.
- **Formal guarantees:** Lean 4 proofs establish authority, vouching, and revocation correctness.
- **Ecosystem integration:** Cross-chain resolution and IAM bridges ensure practical adoption.

Future work includes post-quantum ZK proof systems (integration with K-CHAIN’s ML-DSA keys [6]), credential aggregation across multiple issuers, and reputation scoring based on credential usage patterns.

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